# Bayesian Inference and Hypothesis Testing

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# Outline

Introduction

Conditional Probability and the Theorem of Bayes

Simple Bayesian hypothesis testing

Bayesian Inference

Exercises

Introduction

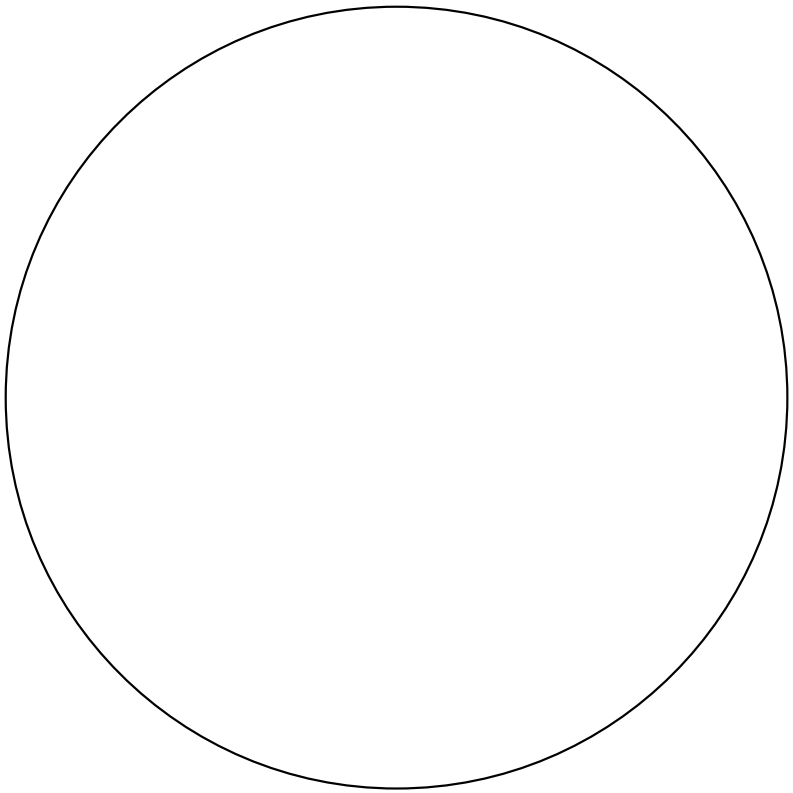
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# Probability refresher

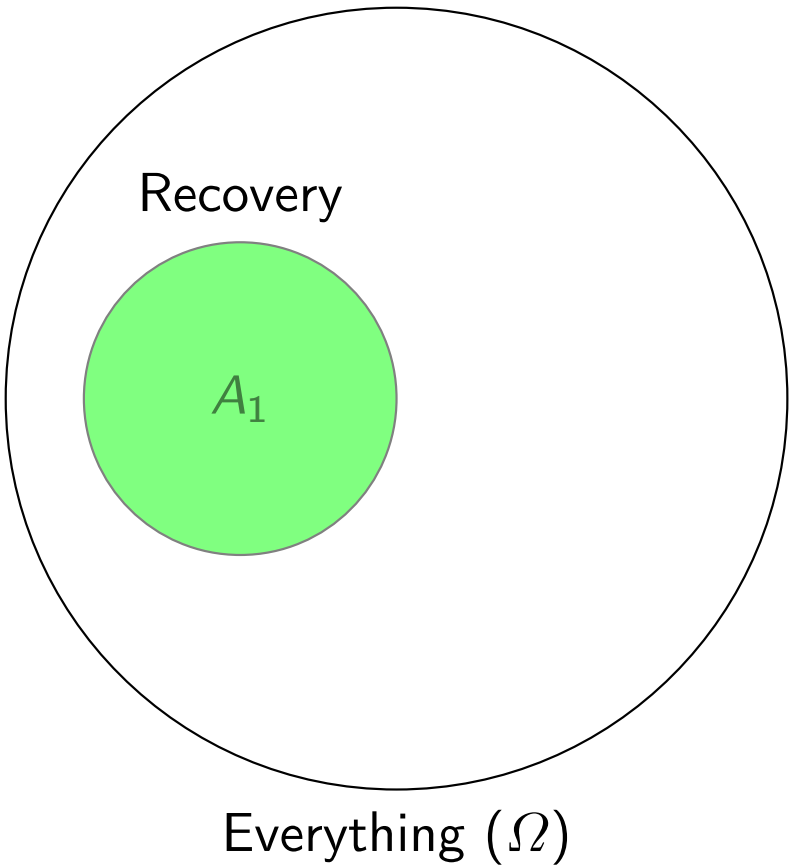


Everything ($\Omega$)

## Notation

- $\omega \in \Omega$ random outcome

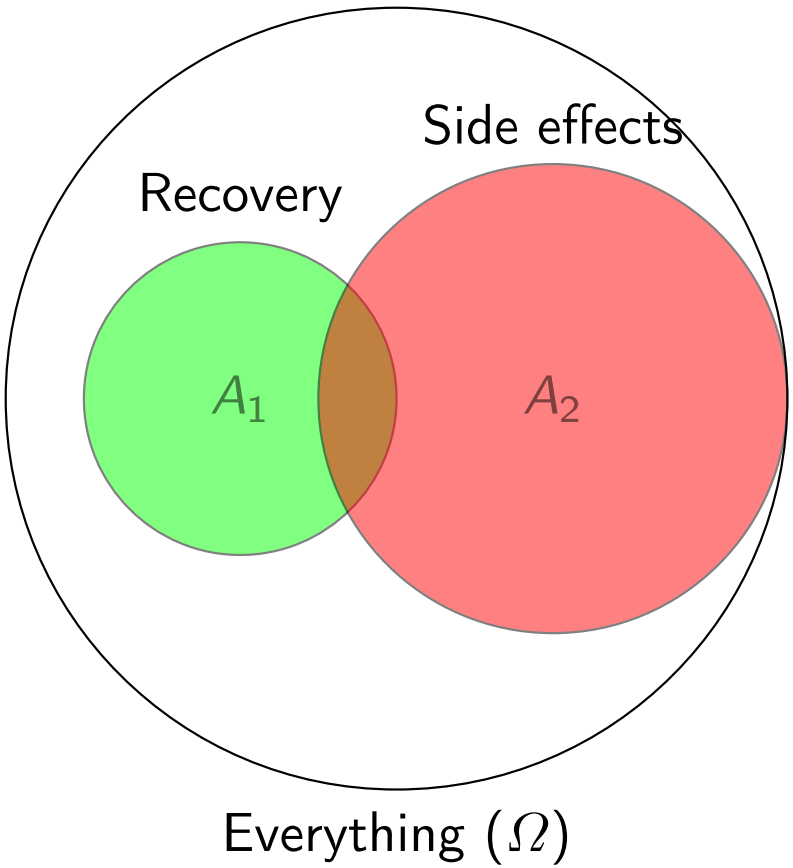
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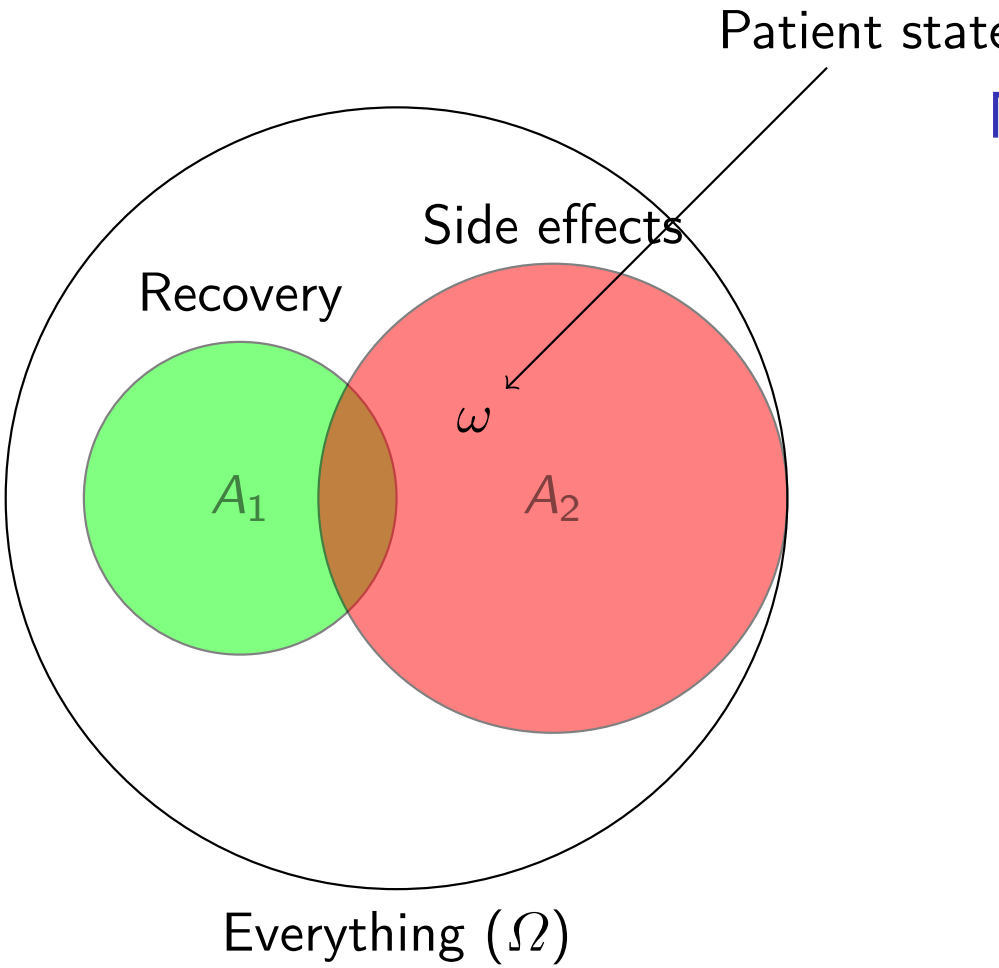
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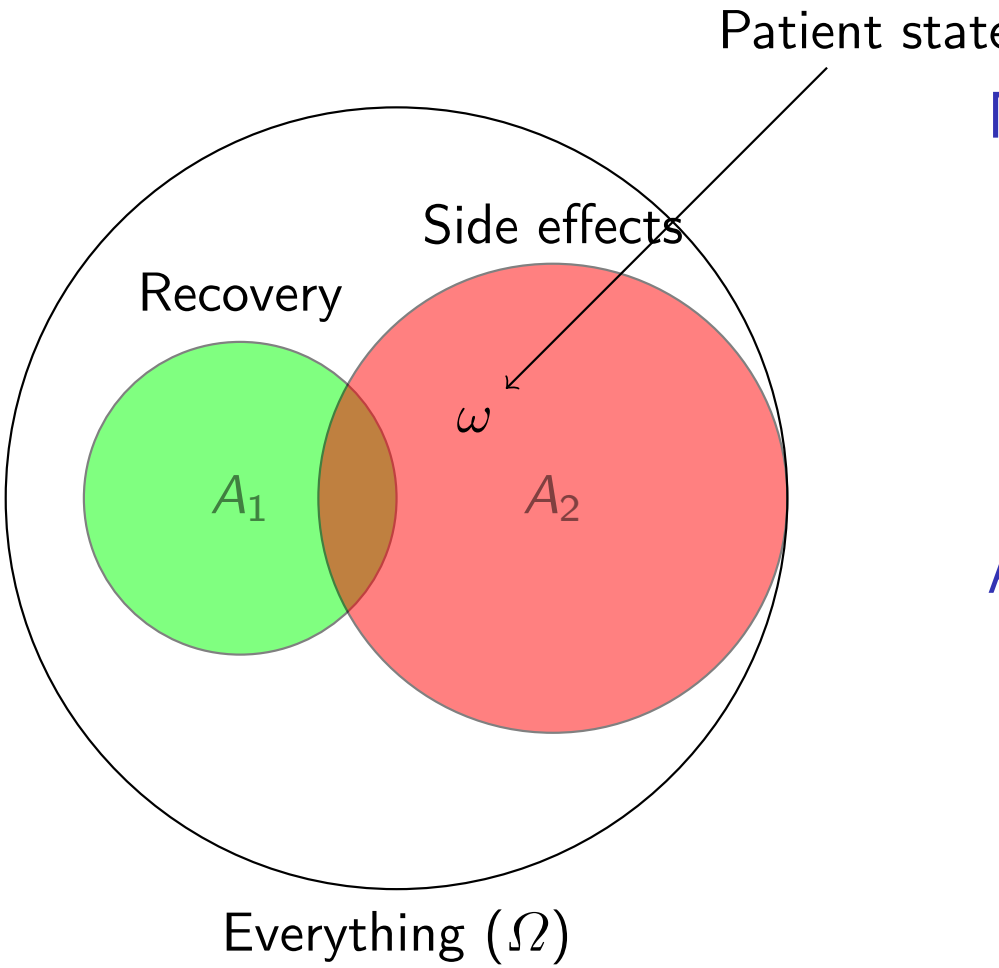
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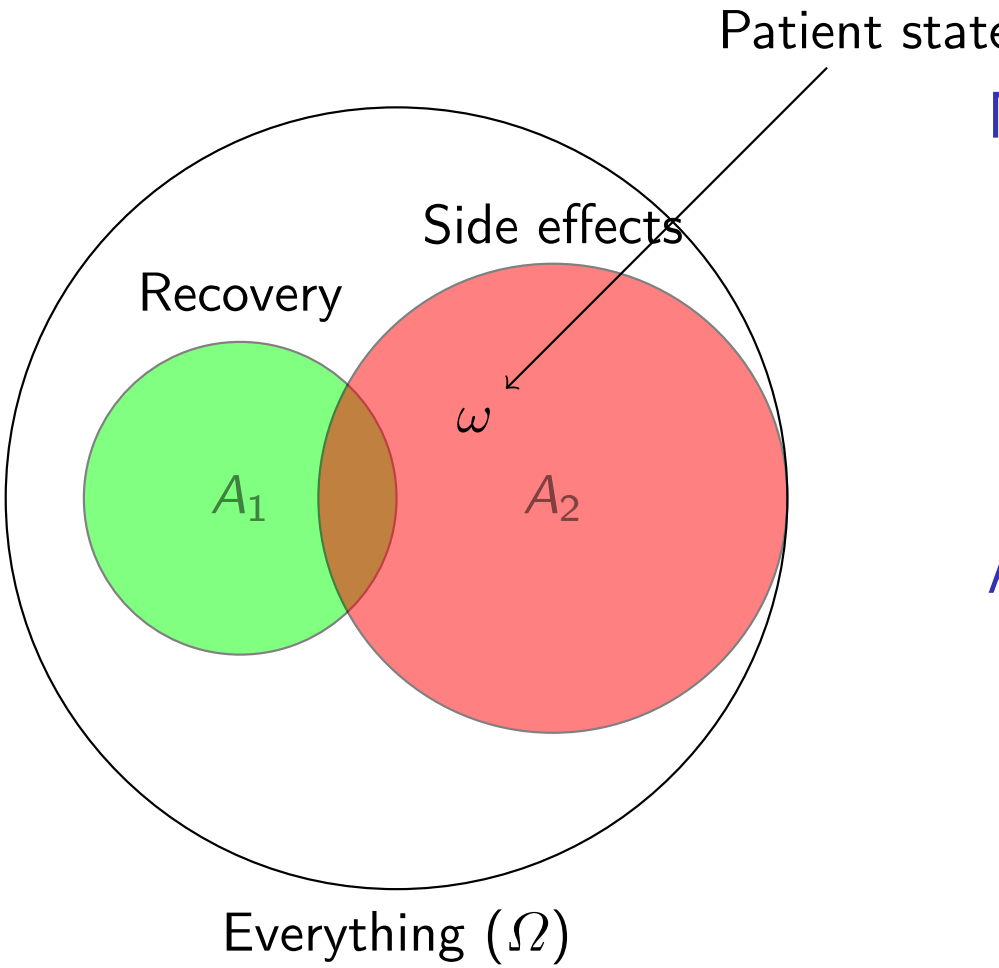

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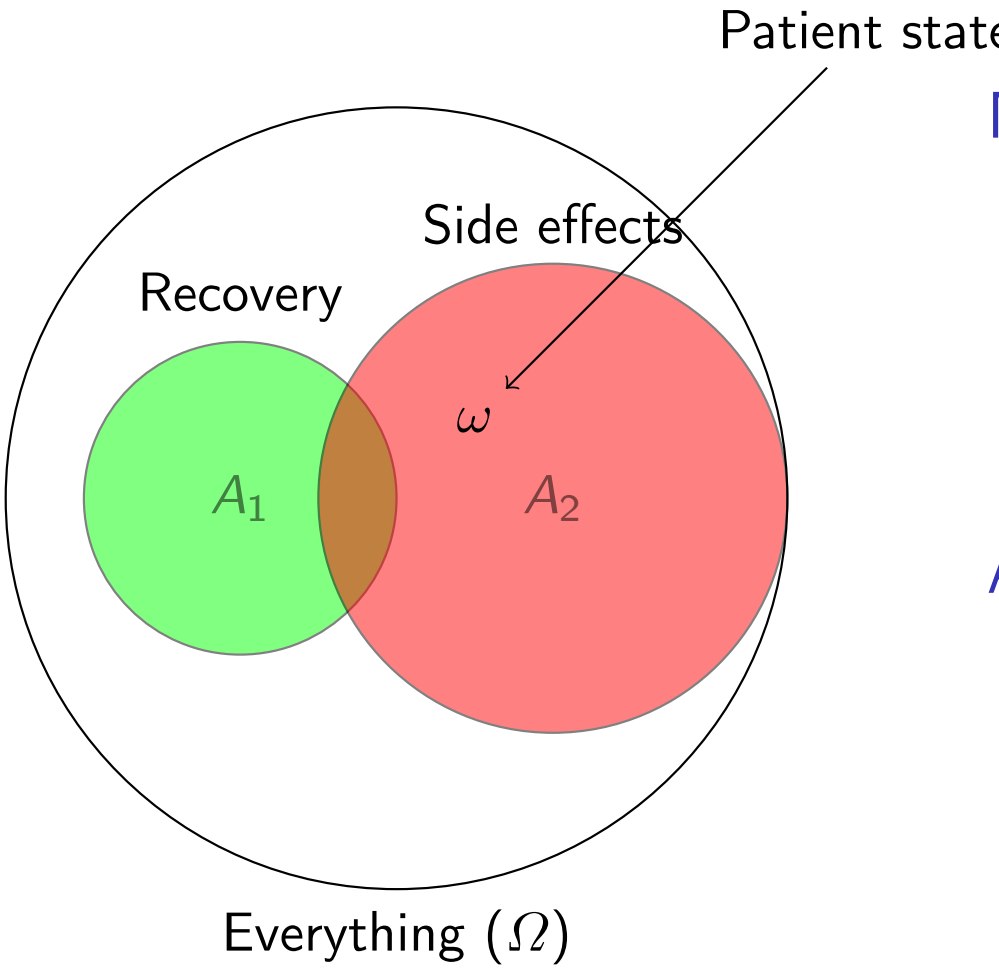

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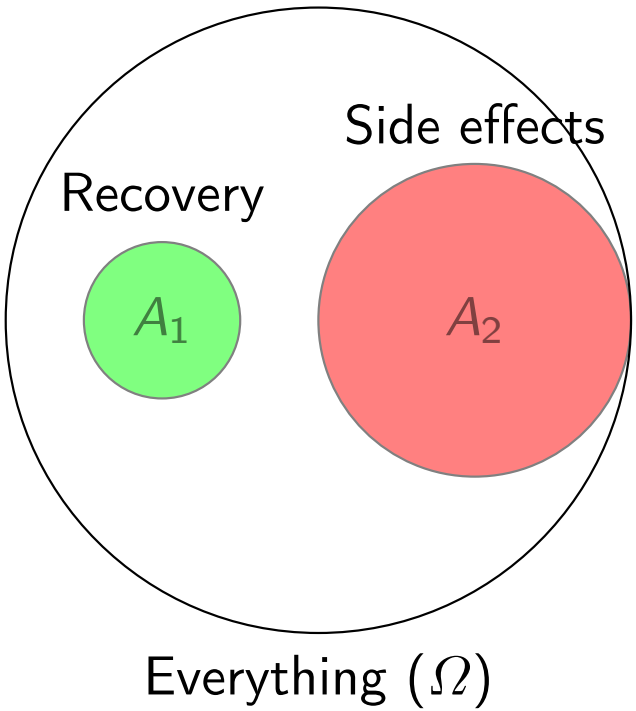

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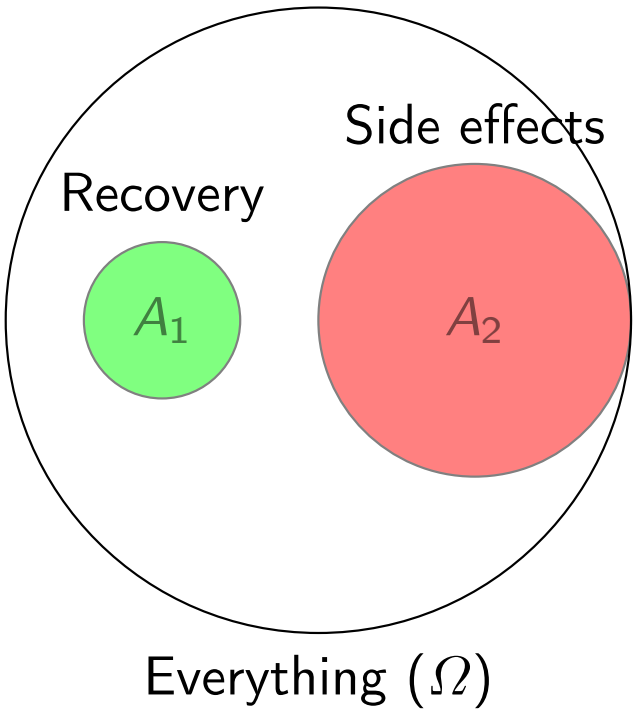
- $P(A) \geq 0$ for any $A \subset \Omega$.
- $P(\Omega) = 1$.
- If $A_1 \cap A_2 = \emptyset$ then $P(A_1 \cup A_2) = P(A_1) + P(A_2)$.

# Mutually exclusive events




- $A_1 \cap A_2 = \emptyset$
- Consequently, $P(A_1 \cup A_2) = P(A_1) + P(A_2)$
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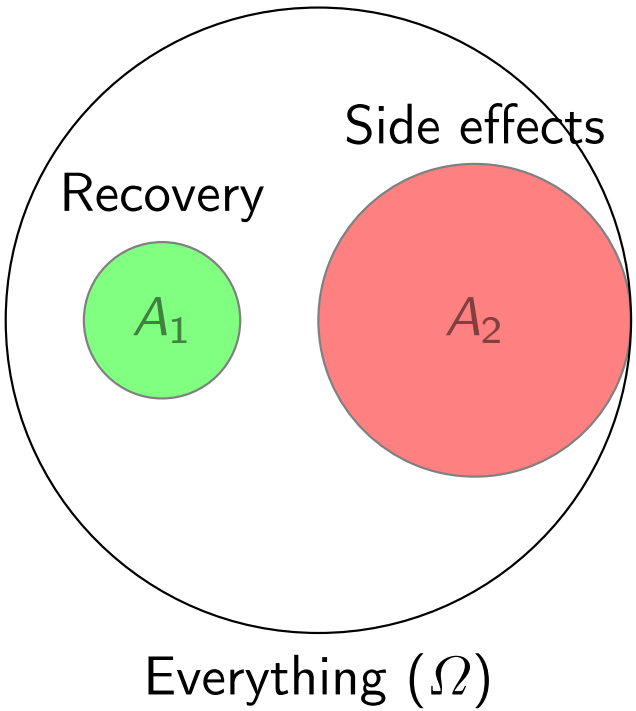



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## Which of the following is true?

1. If $A_2$ happens then $A_1$ happens.
2. If $A_1$ happens then $A_2$ happens.
3. $A_1$ and $A_2$ can happen at the same time.
4. $A_1$ and $A_2$ cannot happen at the same time.

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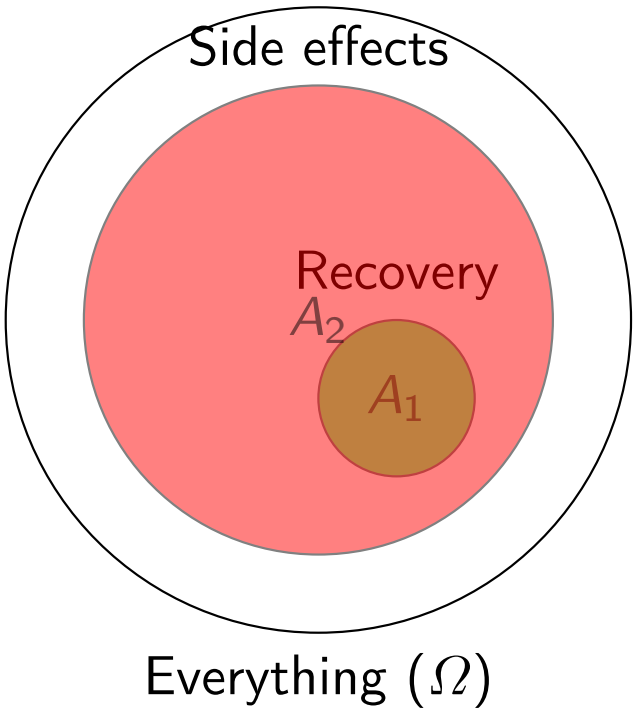

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- (4) is correct.

# Implication

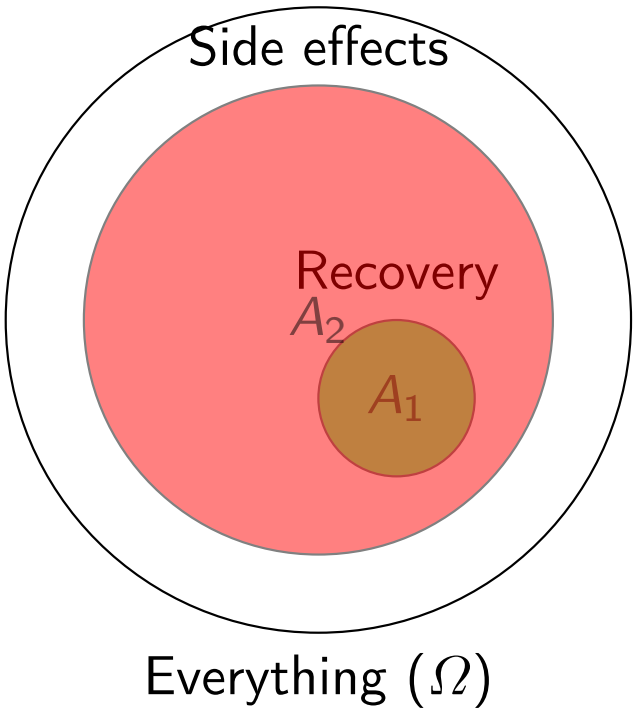



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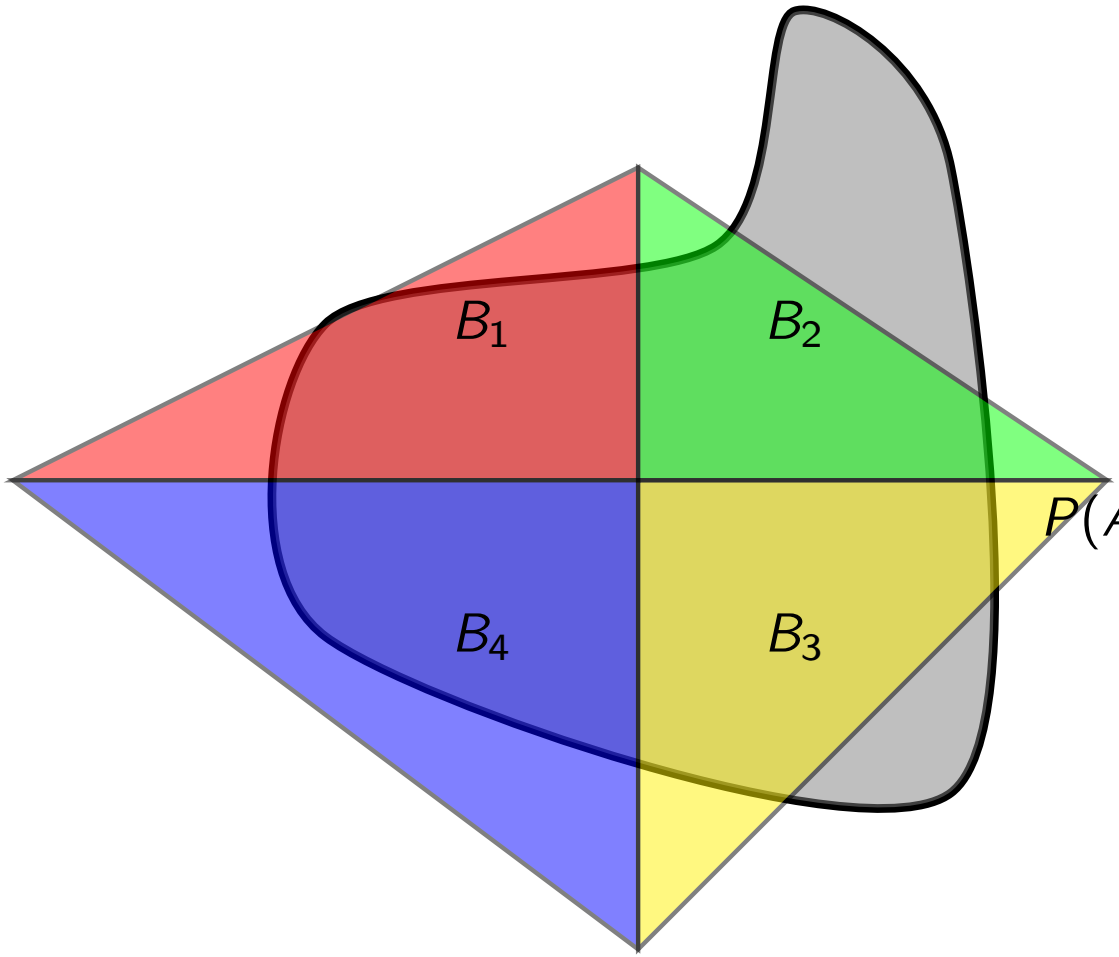

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- (2, 3) are correct

# Marginalisation




$$P(B) = P\left(\bigcup_i B_i\right) = \sum_i P(B_i)$$

$$P(A \cap B) = P\left(\bigcup_i (A \cap B_i)\right)$$
$$= \sum_i P(A \cap B_i),$$

# Exercises

## Implication

- Prove that if $A \subset B$ then $P(A) \leq P(B)$.

## Independent vs mutually exclusive events

- Prove that, if $A, B$ are mutually exclusive events with $P(A) > 0, P(B) > 0$ then they are not independent.

[Reminder: if $A, B$ are independent then $P(A \cap B) = P(A)P(B)$.

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# Bayes theorem

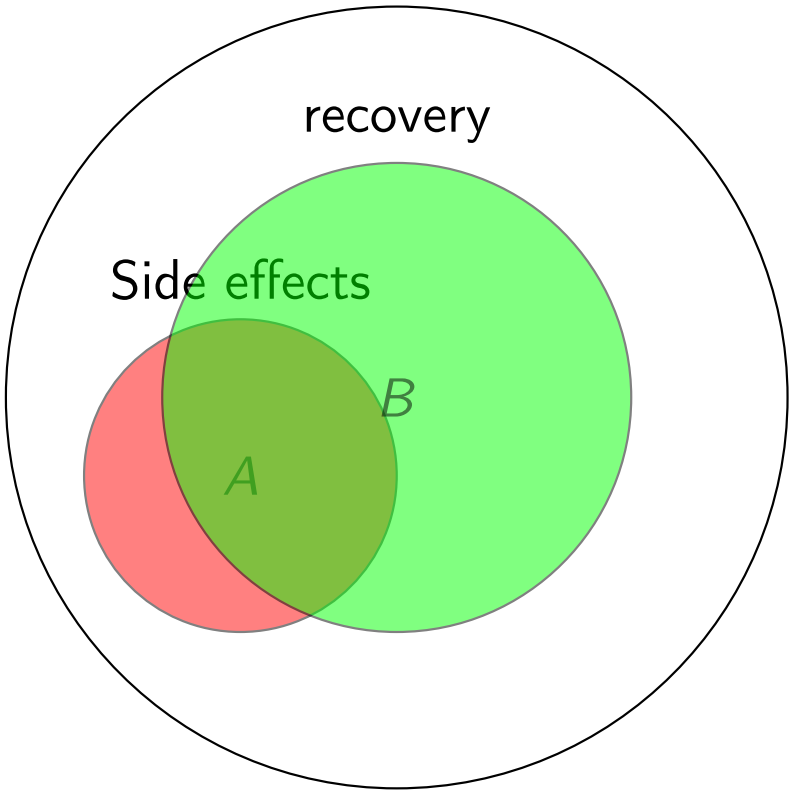
## Conditional probability

The probability of A happening if B happens.

$$P(A|B) \triangleq \frac{P(A \cap B)}{P(B)}$$

Since the following is also true:

$$P(B|A) \triangleq \frac{P(A \cap B)}{P(A)}$$




## Bayes's theorem

So we can reverse the order of conditioning:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

- If you do recover, what is more likely? Side-effects or not?
- If you have side-effects, are you more likely to recover not?

# The cards problem

1. Print out a number of cards, with either [A|A], [A|B] or [B|B] on their sides.
2. If you have an A, what is the probability of an A on the other side?
3. Have the students perform the experiment with:
   - 3.1 Draw a random card.
   - 3.2 Count the number of people with A.
   - 3.3 What is the probability that somebody with an A on one side will have an A on the other?
   - 3.4 Half of the people should have an A?

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## The prior and posterior probabilities

| | | | | |
|---|---|---|---|---|
| A | A | 2/6 | A observed | 2/3 |
| A | B | 1/6 | A observed | 1/3 |
| B | A | 1/6 | | |
| B | B | 2/6 | | |

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- This is a purely subjective measure!

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- The result is either positive or negative ($\neg D$).
- What is your belief now that the suspect is guilty?

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- What is your belief that the people with the positive test are guilty?

# Explanation

- Prior: $P(H_i)$. How much do we believe in $H_i$.
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- Posterior $P(H_i|D)$: How likely is $H_i$ given the data

$$P(H_i|D) = \frac{P(D \cap H_i)}{P(D)} = \frac{P(D|H_i)P(H_i)}{P(D)}$$

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$$P(H_i|D) = \frac{P(D \cap H_i)}{P(D)} = \frac{P(D|H_i)P(H_i)}{P(D)}$$

- Obtaining the marginal probability:

$$P(D) = P(D \cap H_0) + P(D \cap H_1) = P(D|H_0)P(H_0) + P(D|H_1)P(H_1)$$

- Posterior:

$$P(H_0|D) = \frac{P(D|H_0)P(H_0)}{P(D|H_0)P(H_0) + P(D|H_1)P(H_1)}$$

- Assuming $P(D|H_1) = 1$, and setting $P(H_0) = q$, this gives

$$P(H_0|D) = \frac{0.1q}{0.1q + 1 - q} = \frac{q}{10 - 9q}$$

- The posterior can always be updated with more data!

# The Covid Test

- You are feeling sick. Two possibilities:
  - $H_0$: You do not have COVID.
  - $H_1$: You have COVID.
- You perform a rapid test, which has detects COVID 80% of the time.

$$P(T|H_1) = 0.8.$$

- It falsely says you have COVID 10 % of the time.

$$P(T|H_0) = 0.1.$$



- Throw two dice dice and fill in the result in the google sheet to get your test result.
- Select your own prior probability $P(H_1)$ of having covid. What is the posterior probability $P(H_1|T)$

# Python example

```
def get_posterior(prior, data, likelihood):
        marginal = prior*likelihood[data][0]
         + (1 - prior)*likelihood[data][1]
        posterior = prior*likelihood[data][0] / marginal
        return posterior

import numpy as np
prior = 0.9 # Pr(H0)
likelihood = np.zeros([2, 2])
likelihood[0][0] = 0.9 # Pr(F|H0)
likelihood[1][0] = 0.1 # Pr(T|H0)
likelihood[0][1] = 0 # Pr(F|H1)
likelihood[1][1] = 1 # Pr(T|H1)
data = 1
return get_posterior(prior, data, likelihood)
```

# Types of hypothesis testing problems

## Simple Hypothesis Test

Example: DNA evidence, Covid tests

- Two hypothesese $H_0, H_1$
- $P(D|H_i)$ is defined for all $i$

## Multiple Hypotheses Test

Example: Model selection

- $H_i$: One of many mutually exclusive models
- $P(D|H_i)$ is defined for all $i$

## Null Hypothesis Test

Example: Are men's and women's heights the same?

- $H_0$: The 'null' hypothesis
- $P(D|H_0)$ is defined
- The alternative is undefined

# Pitfalls

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## The garden of many paths

- Having a huge hypothesis space
- Selecting the relevant hypothesis after seeing the data

# Class feedback



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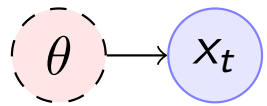
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## Example (Bernoulli model)

- $x \in \{0, 1\}$, $\theta \in [0, 1]$

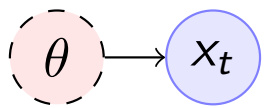


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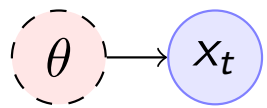


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- $P_\theta(1) = \theta$

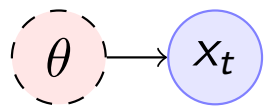


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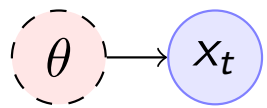


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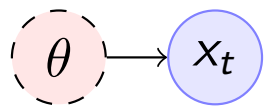


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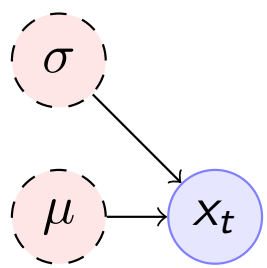
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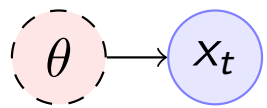


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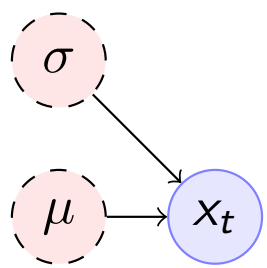
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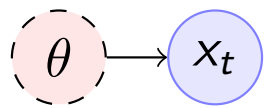


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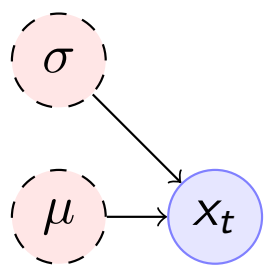
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- $p_\theta(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$



# Maximum likelihood (ML) inference

- Family $\{P_\theta | \theta \in \Theta\}$
- Data $x$ with likelihood $P_\theta(x)$ for each parameter value $\theta$.
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- Show that $\theta_{\mathrm{ML}}(x) = s_T / T$.
- What is the problem with this estimate?

# Gaussian model

- $x_t \in \mathbb{R}$, for $t \in [T]$, $\mu \in \mathbb{R}, \sigma \in \mathbb{R}_+$
- $x_t \mid \mu, \sigma \sim \mathrm{Normal}(\mu, \sigma)$
- Follow the same steps as for the Bernoulli model above — write the log-likelihood, differentiate with respect to each parameter separately, set both derivatives to zero — to show that

$$\hat{\mu}_{\mathrm{ML}}(x) = \frac{1}{T}\sum_{t=1}^{T} x_t, \qquad \hat{\sigma}^2_{\mathrm{ML}}(x) = \frac{1}{T}\sum_{t=1}^{T} \left[x_t - \hat{\mu}_{\mathrm{ML}}(x)\right]^2 .$$

- Does this estimator for $\sigma^2$ have a problem analogous to the Bernoulli one above? (Hint: what happens when $T = 1$?)

# Maximum a posteriori (MAP) inference

- Family $\{P_\theta | \theta \in \Theta\}$
- Data $x$ with likelihood $P_\theta(x)$ for each parameter value $\theta$.
- Prior $\phi(\theta)$.
- $\theta_{\text{MAP}}(x) = \arg\max_\theta P_\theta(x)\phi(\theta)$
- Experiment with the prior for the Bernoulli model.

# Bayesian Inference

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$$\phi(\theta|x) = \frac{P_\theta(x)\phi(\theta)}{\sum_{\theta' \in \Theta} P_{\theta'}(x)\phi(\theta')}, \qquad \text{(finite } \Theta\text{, } \phi \text{ is a probability)}$$

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## Alternative notation for different probability spaces

- The prior $\phi(\theta) = \mathbb{P}(\theta)$ and posterior $\phi(\theta \mid x) = \mathbb{P}(\theta \mid x)$ belief.
- The likelihood $P_\theta(x) = \mathbb{P}(x \mid \theta)$
- The marginal $\mathbb{P}_\phi(x) = \sum_\theta P_\theta(x)\phi(\theta)$.

# Probabilistic machine learning

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## Maximum likelihood approach

- Model selection: $\theta^*_{ML}(x) = \arg\max_\theta P_\theta(x)$.
- Model prediction: $P_{\theta^*_{ML}(x)}(x_{t+1})$

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## Bayesian approach

- Posterior calculation: $\phi(\theta|x) = P_\theta(x)\phi(\theta)/\mathbb{P}_\phi(x)$
- Model prediction: $\mathbb{P}_\phi(x_{t+1}|x) = \sum_\theta P_\theta(x_{t+1})\phi(\theta|x)$

# Differences between approaches

## Maximum likelihood approach

- Ignores model complexity
- Is an optimisation problem

## Maximum a posteriori approach

- Regularises model selection using the prior
- Can be seen as solving the optimisation problem

$$\max_{\theta} \ln P_{\theta}(x) + \ln \phi(\theta),$$

where the prior term $\ln \phi(\theta)$ acts as a regulariser.

## Bayesian approach

- Does not select a single model
- Averages over all models according to their fit and the prior
- Does not result in an optimisation problem.

# The n-meteorologists problem

- Consider $n$ meteorological stations $\{\mu\}$ predicting rainfall.
- $x_t \in \{0, 1\}$ with $x_t = 1$ if it rains on day $t$.
- We have a prior distribution $\phi(\mu)$ for each station.
- At time $t$, station $\mu$ makes as a prediction $P_\mu(x_{t+1}|x_1, \ldots, x_t)$
- We observe $x_{t+1}$ and calculate the posterior $\phi(\mu|x_1, \ldots, x_t, x_{t+1})$.

## The marginal distribution

To take into account all stations, we can marginalise:

$$\mathbb{P}_\phi(x_{t+1} \mid x_1, \ldots x_t) = \sum_\mu P_\mu(x_{t+1}|x_t)\phi(\mu)$$

## The posterior

- Show that

$$\phi(\mu \mid x_1, \ldots, x_{t+1}) = \frac{P_\mu(x_t \mid x_1, \ldots, x_t)\phi(\mu|x_1, \ldots, x_t)}{\sum_{\mu'} P_{\mu'}(x_t \mid x_1, \ldots, x_t)\phi(\mu'|x_1, \ldots, x_t)}$$

- How would you implement an ML or a MAP solution to this problem?

# Sufficient statistics

## A statistic $f$

This is any function $f : X \to S$ where

- $X$ is the data space
- $S$ is an arbitrary space

## Example statistics for $X = \mathbb{R}^*$ (the set of all real-valued sequences)

- The sample mean of a sequence $1/T \sum_{t=1}^{T} x_t$
- The total number of samples $T$

## Sufficient statistic

$f$ is sufficient for a family $\{P_\theta : \theta \in \Theta\}$ when

$$f(x) = f(x') \Rightarrow P_\theta(x) = P_\theta(x') \forall \theta \in \Theta.$$

If there exists a finite-dimensional sufficient statistic, Bayesian and ML learning can be done in closed form within the family.

# Conjugate priors

Consider a parametrised family of priors $\Phi$ on $\Theta$ and a distribution family $\{P_\theta\}$
The pair is conjugate if, for any prior $\phi \in \Phi$, and any observation $x$, there exists $\phi' \in \Phi$ such that $\phi'(\theta) = \phi(\theta|x)$

## Standard Parametric conjugate families

| Prior | Likelihood | Parameters $\theta$ | Observations $x$ |
|---|---|---|---|
| Beta | Bernoulli | $[0, 1]$ | $\{0, 1\}^T$ |
| Multinomial | Dirichlet | $\mathbb{\Delta}^n$ | $\{1, \ldots, n\}^T$ |
| Gamma | Normal | $\mathbb{R}, \mathbb{R}$ | $\mathbb{R}^T$ |
| Wishart | Normal | $\mathbb{R}^n, \mathbb{R}^{n\times n}$ | $\mathbb{R}^{n\times T}$ |

The Simplex $\mathbb{\Delta}^n = \{\boldsymbol{\theta} \in [0, 1]^n : \|\boldsymbol{\theta}\|_1\}$ is the set of all $n$-dimensional probability vectors.

## Extensions

- Discrete Bayesian Networks.
- Linear-Gaussian Models (i.e. Bayesian linear regression)
- Gaussian Processes.

# Beta-Bernoulli




## Definition of the Bernoulli distribution

If $x_t \mid \theta \sim \text{Bernoulli}(\theta)$. $\theta \in [0, 1]$, $x_t \in \{0, 1\}$ and:

$$P_\theta(x_t = 1) = \theta$$

## Definition of the Beta density

If $\theta \sim \text{Beta}(\alpha_1, \alpha_0)$, $\alpha_0, \alpha_1 > 0$ and

$$p(\theta|\alpha_1, \alpha_0) \propto \theta^{\alpha_1 - 1}(1 - \theta)^{\alpha_0 - 1}$$

## Beta-Bernoulli conjugate pair

- $\theta \sim \text{Beta}(\alpha_1, \alpha_0)$.
- $x_t \mid \theta \sim \text{Bernoulli}(\theta)$.

Then, for any $x = x_1, \ldots, x_T$, the posterior distribution is

- $\theta \mid x \sim \text{Beta}(\alpha_1 + \sum_t x_t, \alpha_0 + T - \sum_t x_t)$.

# Dirichlet-Multinomial




## Definition of the Multinomial distribution

If $x_t \mid \boldsymbol{\theta} \sim \mathrm{Mult}(\boldsymbol{\theta})$, with $\theta \in \mathbb{\Delta}^n$ and $x_t \in \{1, \ldots, n\}$ and:

$$P_{\boldsymbol{\theta}}(x_t = i) = \theta_i$$

## Definition of the Dirichlet density

If $\boldsymbol{\theta} \sim \mathrm{Dir}(\boldsymbol{\alpha})$, with $\boldsymbol{\alpha} \in \mathbb{R}_+^n$ then $p(\theta|\boldsymbol{\alpha}) \propto \prod_i \theta_i^{\alpha_i - 1}$

## Dirichlet-Multinomial conjugate pair

- $\theta \sim \mathrm{Dir}(\boldsymbol{\alpha})$.
- $x_t \mid \theta \sim \mathrm{Bernoulli}(\boldsymbol{\theta})$.

Then, for any $x = x_1, \ldots, x_T$, the posterior distribution is

- $\theta \mid x \sim \mathrm{Dir}(\boldsymbol{\alpha} + \boldsymbol{s_T})$, where $s_{T,i} = \sum_{t=1}^{T} \mathbb{I}\{x_t = i\}$,

Introduction

Conditional Probability and the Theorem of Bayes

Simple Bayesian hypothesis testing

Bayesian Inference

Exercises

# Probability

## Normal distribution

- Find the MLE for the normal distribution. What is the sufficient statistic?